AENL226 Power Electronics and power systems
DC-DC converter part 1
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Content

- DC-DC converters: Buck converter
- Buck converter with non-idealities
- Simulation example

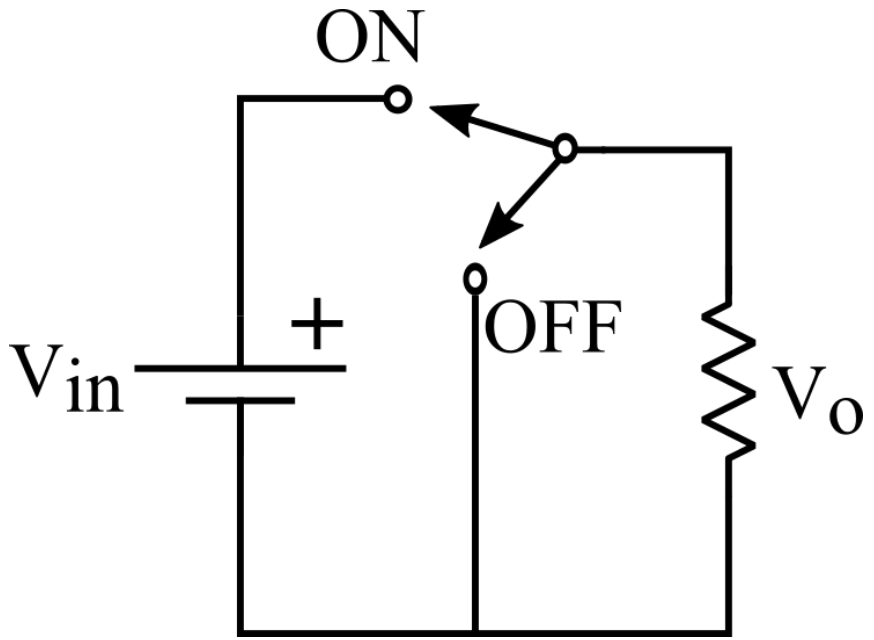
Switching power converter

- DC-DC converters are used for converting DC voltage of a certain magnitude to another magnitude.
- They are usually done by switching regulators rather than linear regulators to improve the efficiency.
- Basic switching regulators include buck, boost and buck boost converters.

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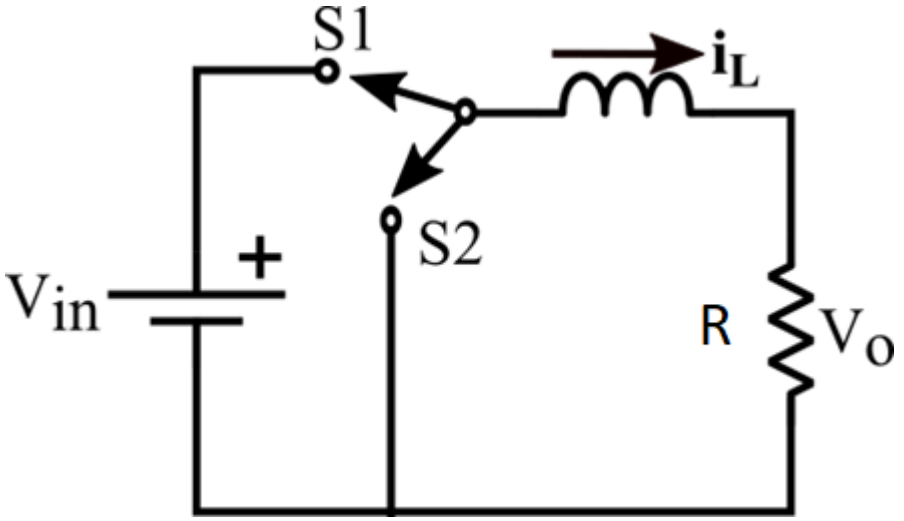
Switching power converter

- In this circuit, the efficiency is 100% however, the output voltage is pulsating.



Switching power converter

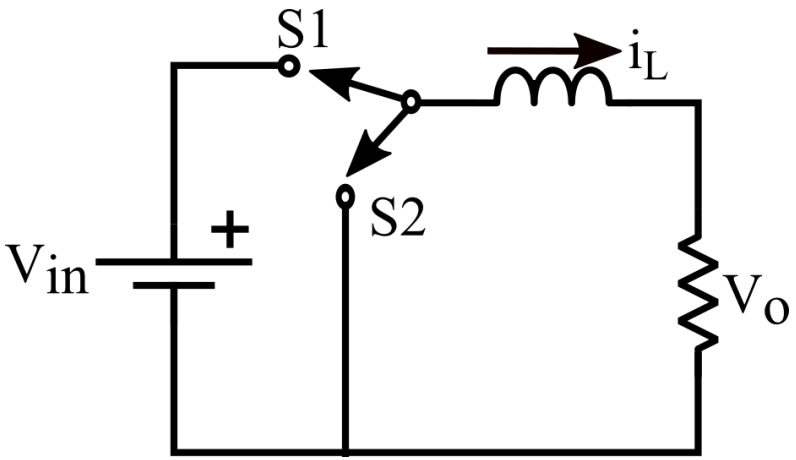
- To provide a smoother output voltage i.e., with less ripple, an inductor is provided.
- The inductor charges and discharges periodically with (L/R) time constant.

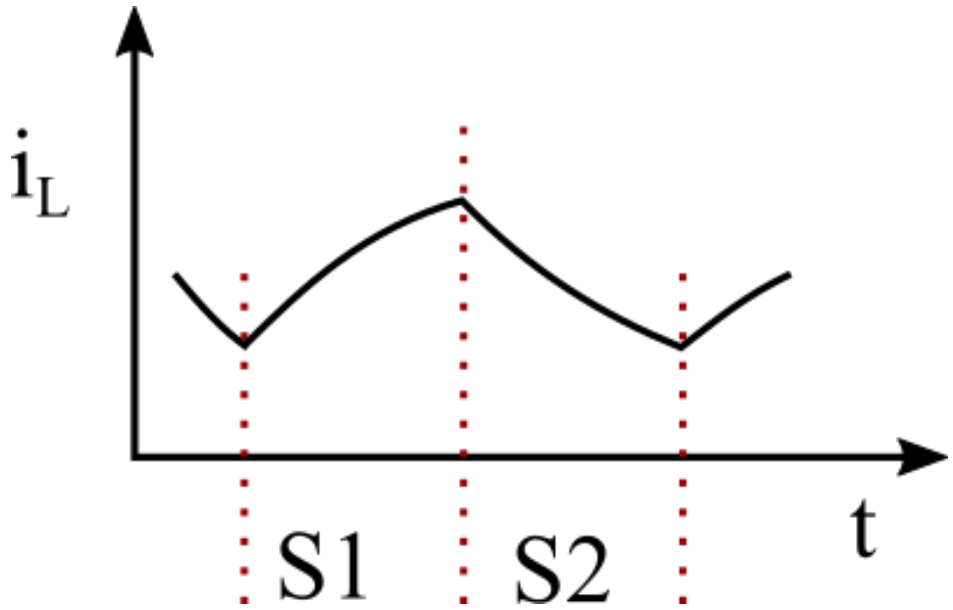


- When switch at S1
  - $v_{in} = L\frac{d}{dt}i_L + Ri_L$
  - $i_L = I_o + \frac{v_{in}}{R}(1 - e^{-t/\tau})$
  - When switch at S2
  - $0 = L\frac{d}{dt}i_L + Ri_L$
  - $i_L = I'_o e^{-t/\tau}$

Switching power converter

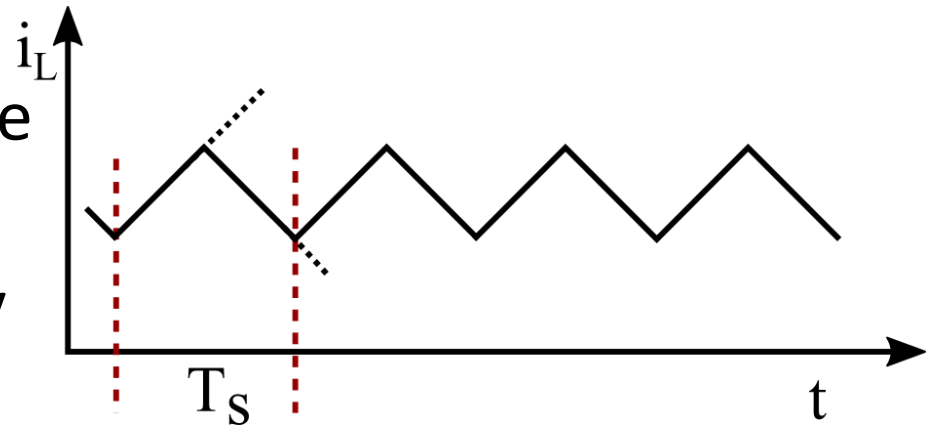
- The inductor charges and discharges periodically with (L/R) time constant.





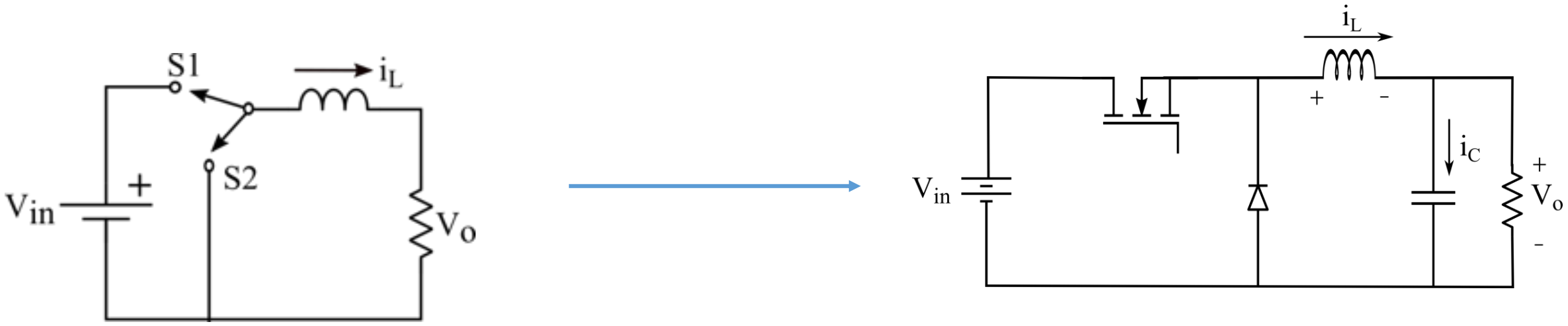
Switching power converter

- If the time constant is much higher than the switching period, then the inductor current can be assumed linear.
- In such a small switching period the ripple in the inductor current can be assumed to be small.
- Initial build up of the load current is possible by having longer duty ratio at the beginning.

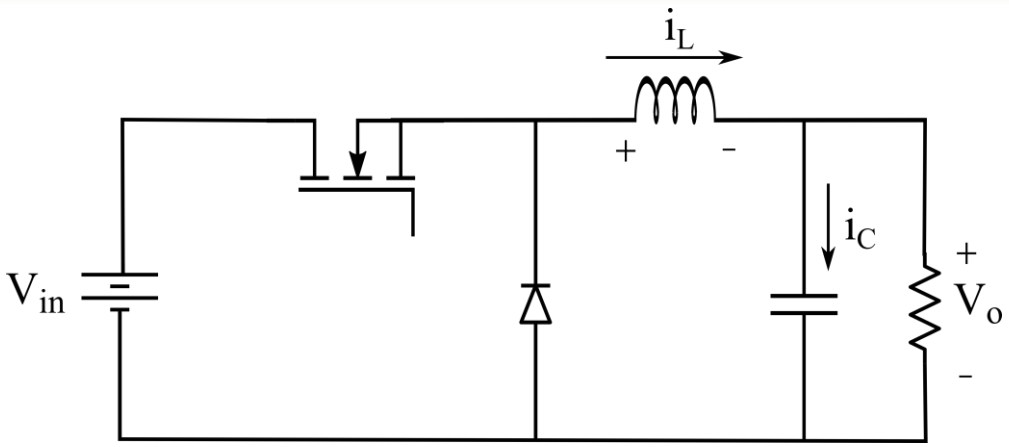


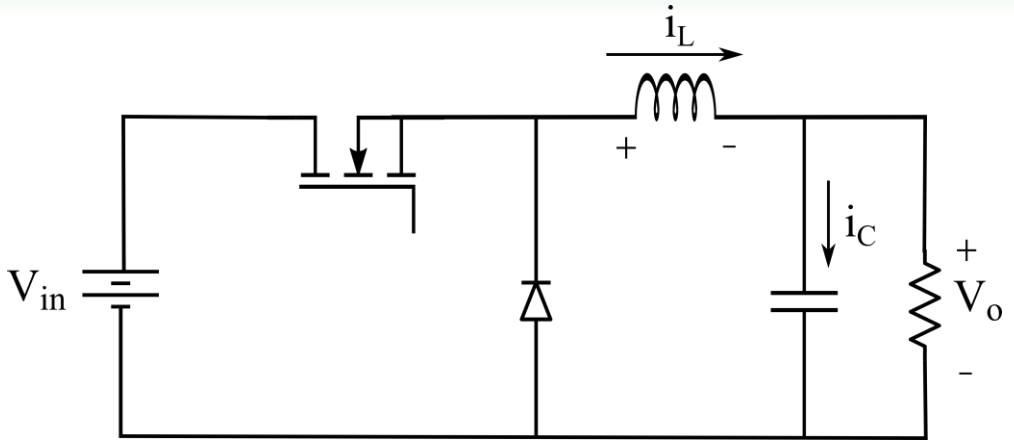
Buck converter

- The output voltage ripple can be further minimized by adding a capacitor at the output.



Inductor currents

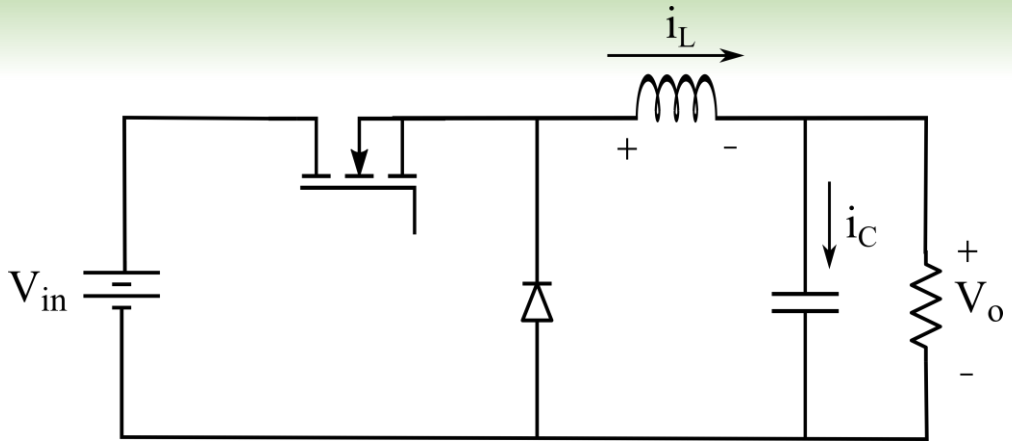




- When switch is on
  - $v_{in} = v_L + v_o$
  - $L\frac{d}{dt}i_L = (v_{in} - v_o)$
  - $\Delta i_L = \frac{(v_{in} - v_o)DT_s}{L}$
- When switch is off
  - $0 = v_L + v_o$
  - $L\frac{d}{dt}i_L = -v_o$
  - $\Delta i_L = \frac{(-v_o)(1-D)T_s}{L}$

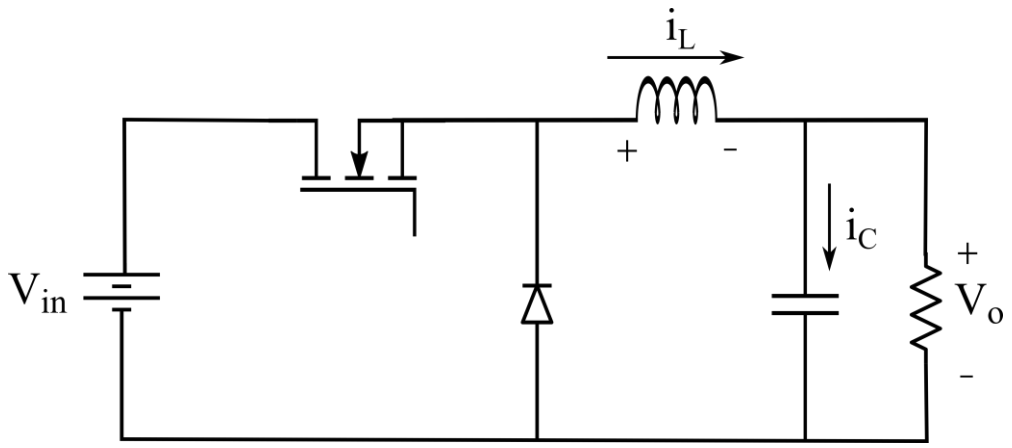
Inductor current

- When steady state is reached, then inductor current buildup over a period of time should be zero.
- The period of time (Ts) is usually taken as the switching time period.
- In other words, at steady state, energy content in the inductor should remain constant in a time period Ts. Similarly, energy content in the capacitor should remain constant in Ts.



Volt second balance

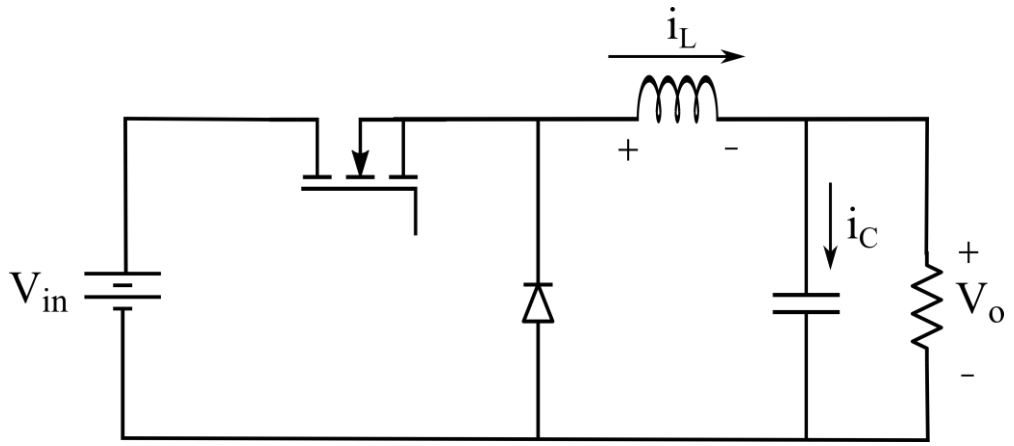
- Mathematically, at steady state, $\int_0^{T_s} di_L = 0 = \frac{1}{L}\int_0^{T_s} v_L \ dt$
- This is called the volt second balance principle on the inductor.
- Similarly, ampere second balance principle on the capacitor (or, any other energy storage element) also holds true i.e., $\int_0^{T_s} dv_c = 0 = \frac{1}{C}\int_0^{T_s} i_c \ dt$



Volt second balance

- The volt second balance gives us the input output relations at steady state.
- We note that, the input and output voltage ripple is usually negligibly small (about 1%) and this assumption can considerably simplify the analysis.

- We note that,
  - $v_{in} - v_o = v_L$ when S1 is ON
  - $-v_o = v_L$ when S2 is ON.
- With negligible ripple voltage,
  - $V_{in} - V_o = v_L$ when S1 is ON
  - $-V_o = v_L$ when S2 is ON.



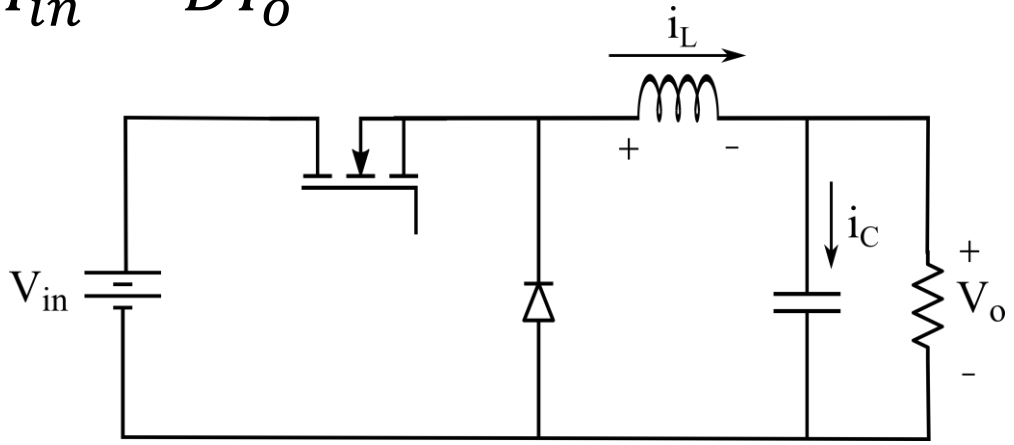
Input output voltage relation

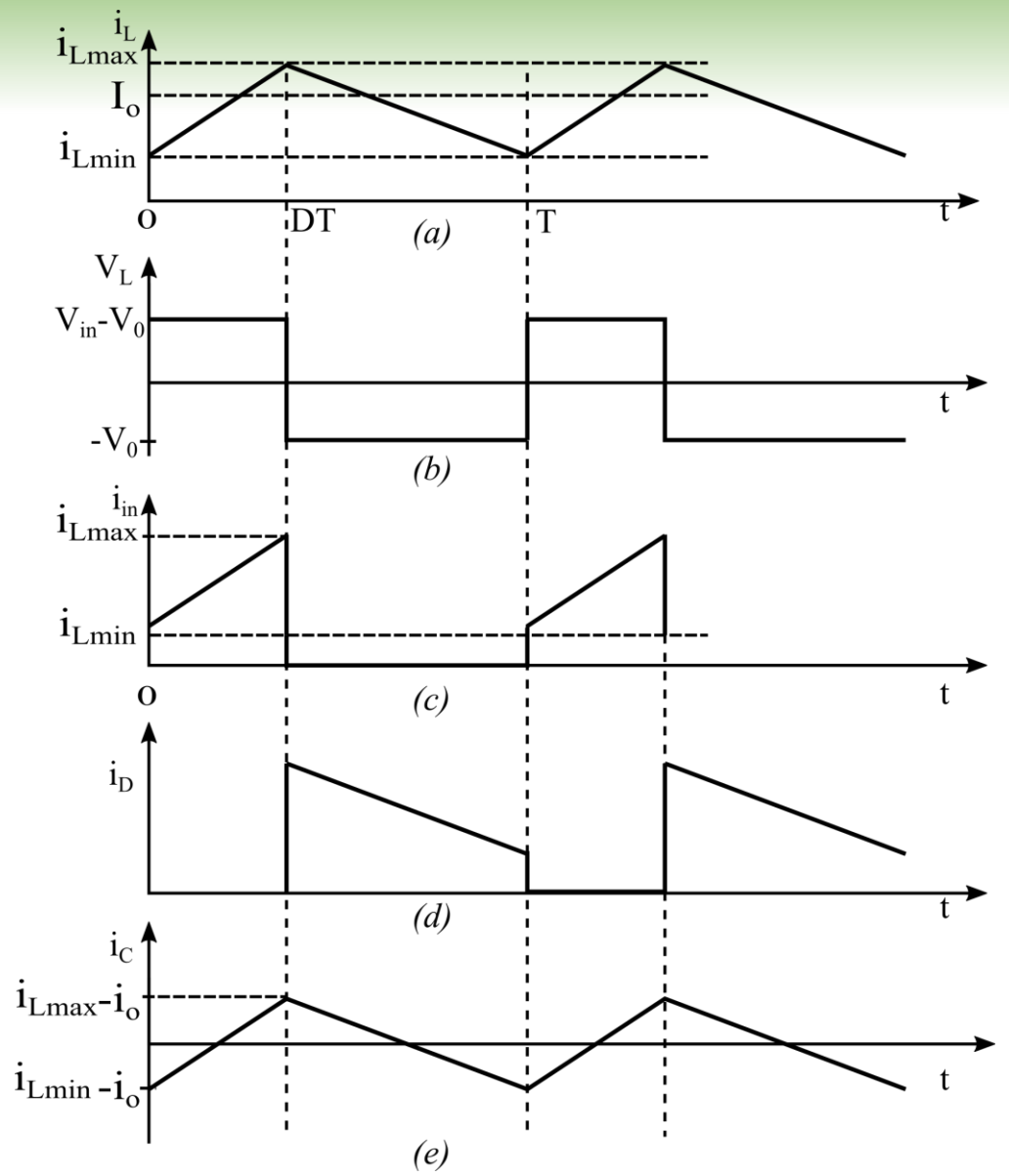
- Thus,
- $0 = \int_0^{T_s} v_L \ dt = (V_{in} - V_o)DT_s + (-V_o)(T_s - DT_s)$

- On simplification, $V_o = DV_{in}$
- This the steady state input output voltage equation of the buck converter.
- Volt second balance gives us a technique to find the input output relation in any converter. We will use it also in other converters.

Waveforms

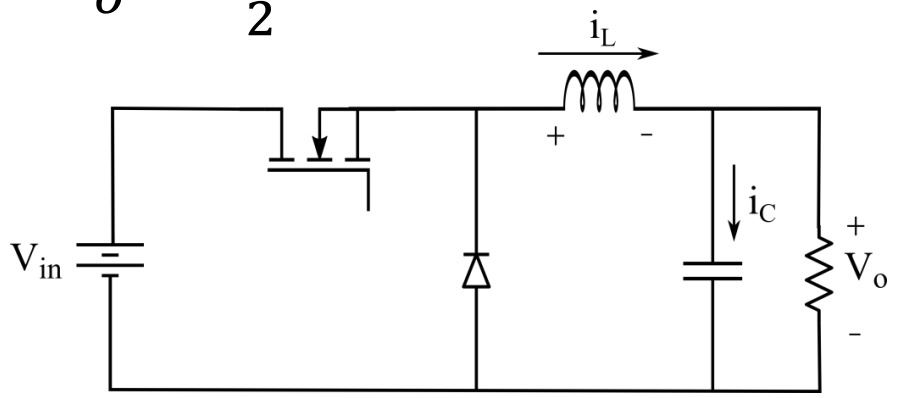
- The other important waveforms for the buck converter can be obtained once the input output relations are obtained.
- E.g., since $V_o I_o = V_{in} I_{in}$, average input current can be obtained: $I_{in} = D I_o$

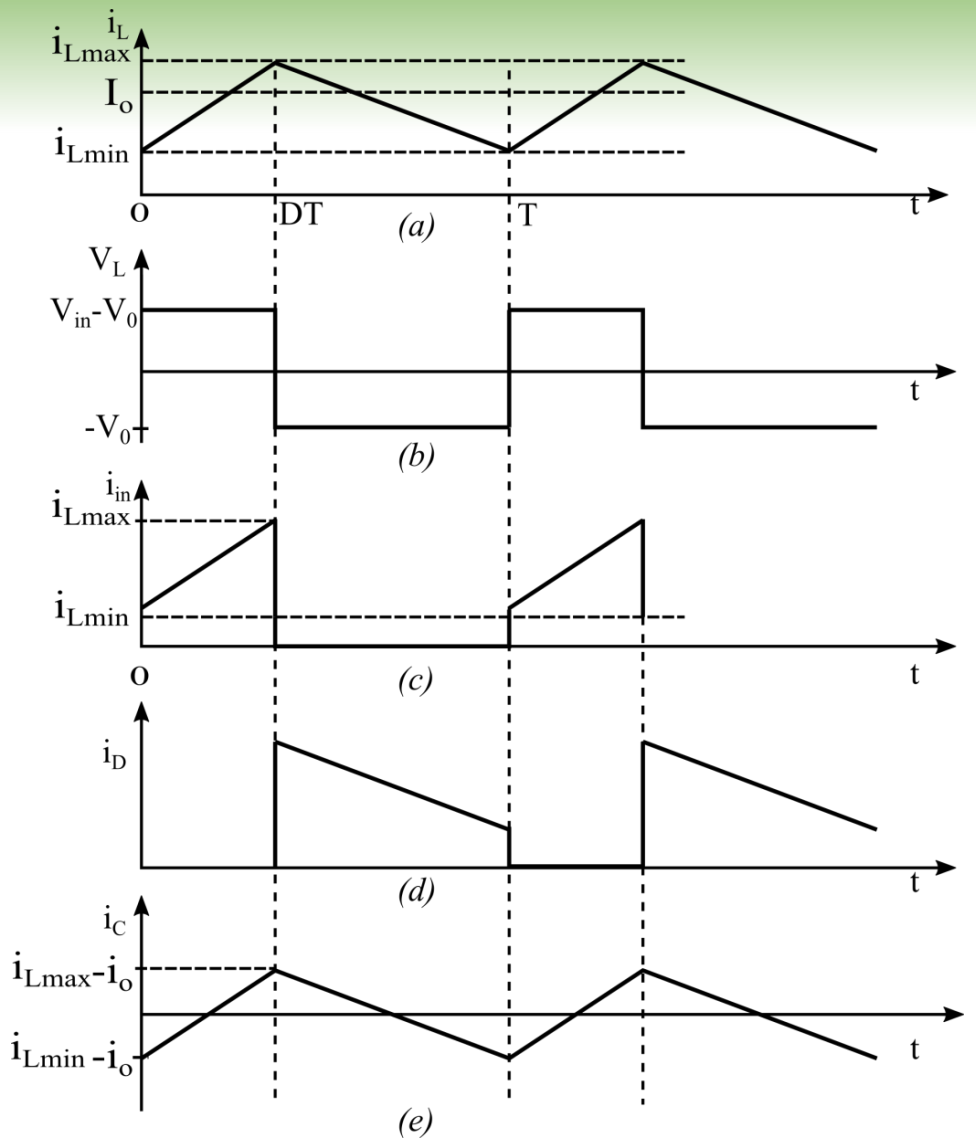




Relationships

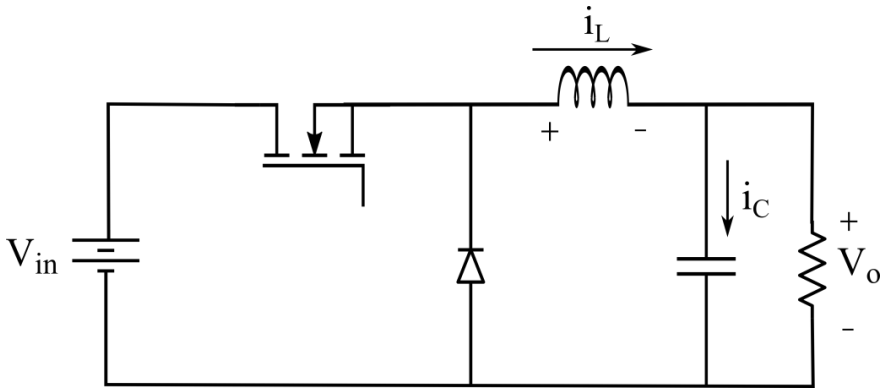
- Several relationships can be derived e.g., inductor current maximum and minimum values.
- Inductor ripple current $\Delta i_L = \left| \frac{(-V_o)(1-D)T_S}{L} \right|$
- $I_{Lmin} = I_o - \frac{\Delta i_L}{2}$
- $I_{Lmax} = I_o + \frac{\Delta i_L}{2}$

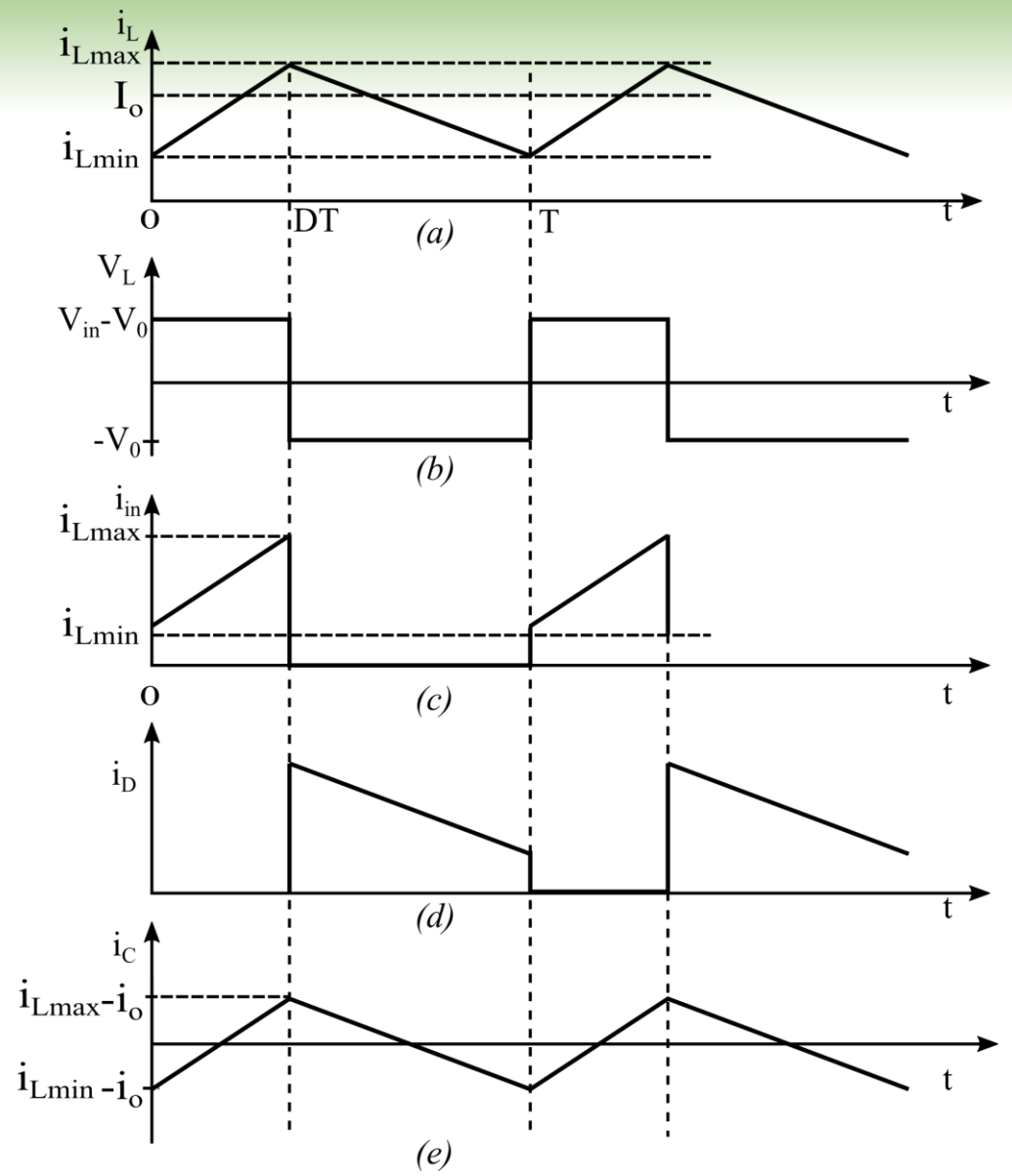




Waveforms

- Average value of inductor current = load current.
- Average current through capacitor=0. Average value of inductor voltage=0.
- Switching frequency of switch and diode are same.





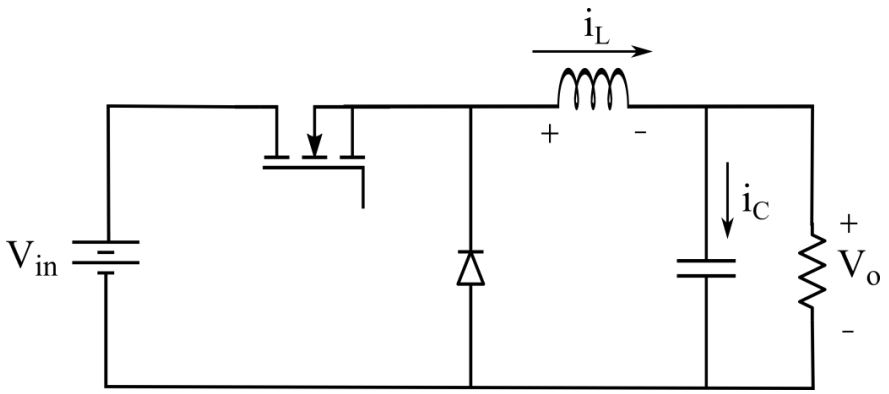
Design of L

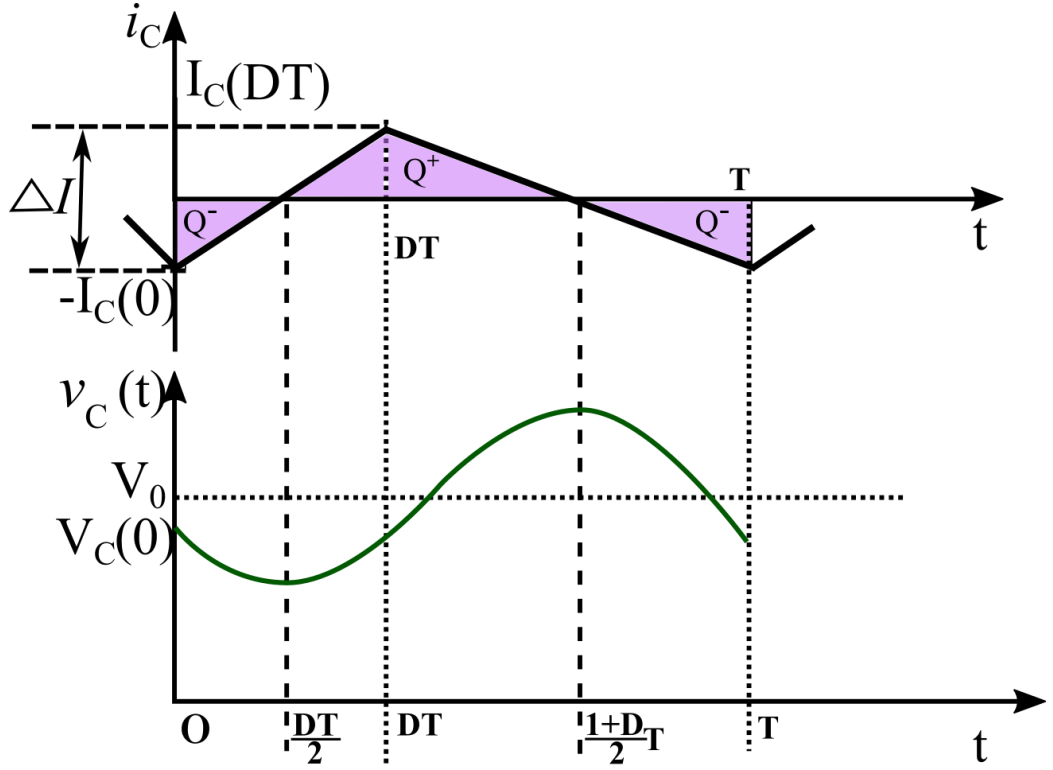
- Usually, the inductor ripple is taken as 10% of the inductor current.
- From the equation $\Delta i_L = \frac{(-V_o)(1-D)T_s}{L}$, taking absolute values,
- $L = \frac{V_o\,(1-D)}{\Delta i_L f_s}$

- Size of inductor is inversely proportional to switching frequency.
- If $\Delta i_L, f_s, V_o$ are constant, then $L = \frac{V_o\,(1-D)}{\Delta i_L f_s}$

Design of C

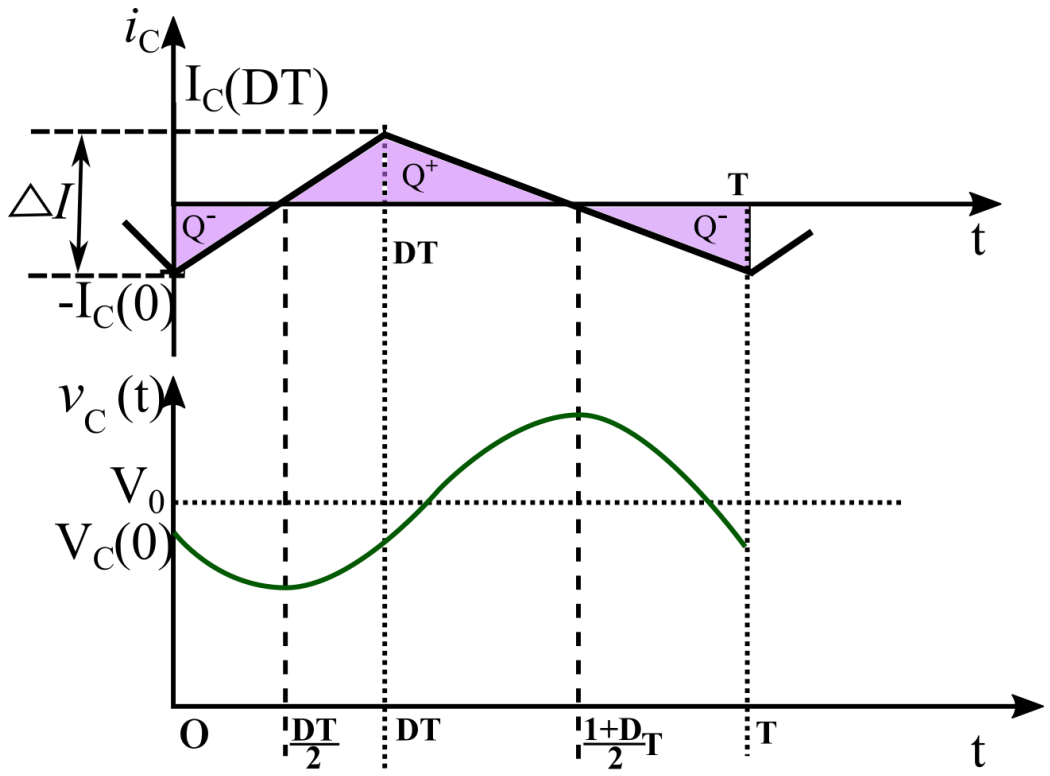
- The capacitance is calculated using charge balance (or Ampere second) method.
- The DC flows through the load, while the ripple flows through the capacitor.





Design of C

- The capacitor voltage ripple is usually taken 1-2% of the capacitor voltage.
- $\Delta Q = C\Delta v_o = \frac{1}{2}\frac{T_s}{2}\frac{\Delta i_L}{2}$
- $C = \frac{\Delta i_L}{8 f_s \Delta v_o}$
- Since $v_c = \int i_c \, dt$, and the capacitor current is triangular, so the capacitor ripple voltage is parabolic.
- Size of capacitor is inversely proportional to switching frequency.
- By substitution, $\frac{\Delta v_o}{V_o} = \frac{(1-D)}{8LCf_s^2}$



Observation for Buck converter

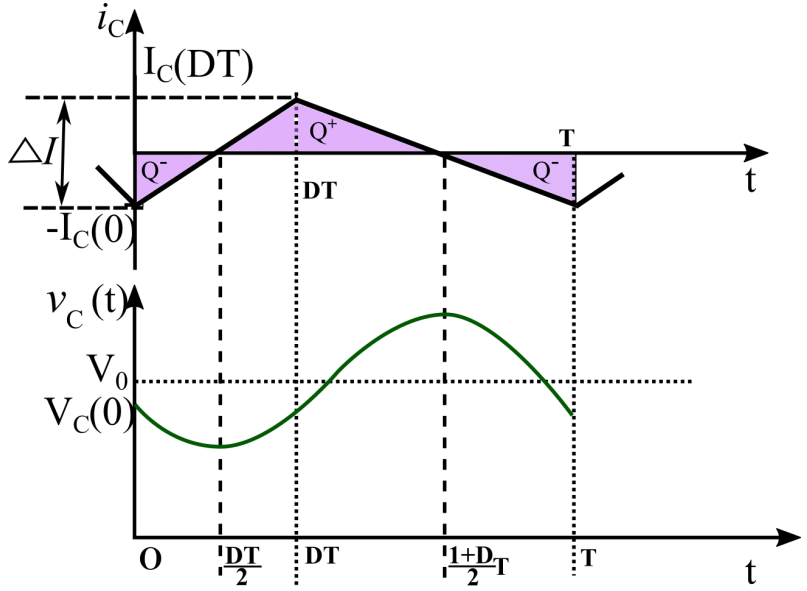
- The output voltage is always less than input voltage. The voltage gain of the converter is independent of switching frequency.
- The output voltage ripple is independent of the load resistance.
- The input current is discontinuous and pulsating. If the source cannot supply such a current, then we should provide an input filter.
- The ideal efficiency is 1, however practical efficiency is less than 1 (e.g., 0.85 to 0.95).

Example

- A speaker in a car consumes about 45W at 15V for which a buck converter is used. The buck converter steps down an input voltage of 24V to 15V with a load current of 3A. What are the maximum and minimum values of inductor currents, and % capacitor ripple voltage, if L=750 uH, C= 1.5 uF . Consider switching frequency of 50 kHz.
- $D = \frac{15}{24} = 0.625.$
- $R = \frac{15}{3} = 5\ \Omega$
- $I_{L(avg)} = I_{load} = 3\ A$
- $L\frac{d}{dt}i_L = (V_{in} - V_o)$. Hence, $\Delta i_L = \frac{(V_{in} - V_o)DT_s}{L} = 0.15\ A$
- Thus, $I_{Lmin} = 2.925\ A$ and $I_{Lmax} = 3.075\ A$

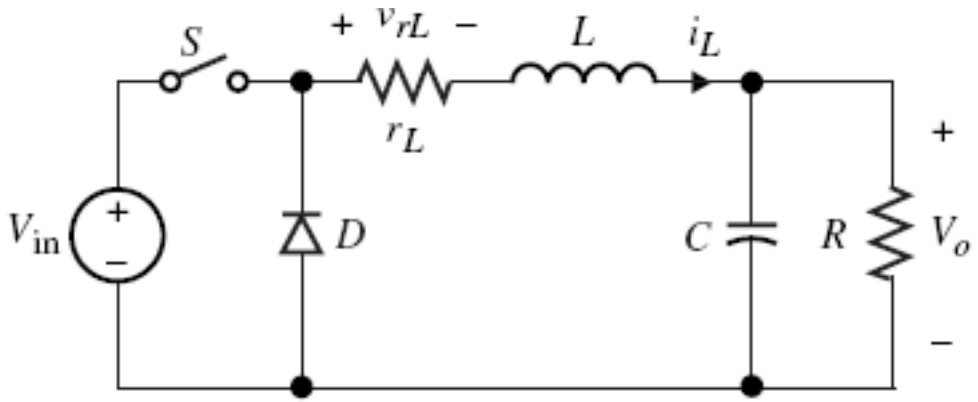
Example

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- $\Delta Q = C\Delta v_o = \frac{1}{2}\frac{T_s}{2}\frac{\Delta i_L}{2}$
- Thus, $\Delta v_o = \frac{0.15}{8\times1.5\times10^{-6}\times50\times10^{3}} = 0.25V$
- Hence % ripple voltage = 1.6%.



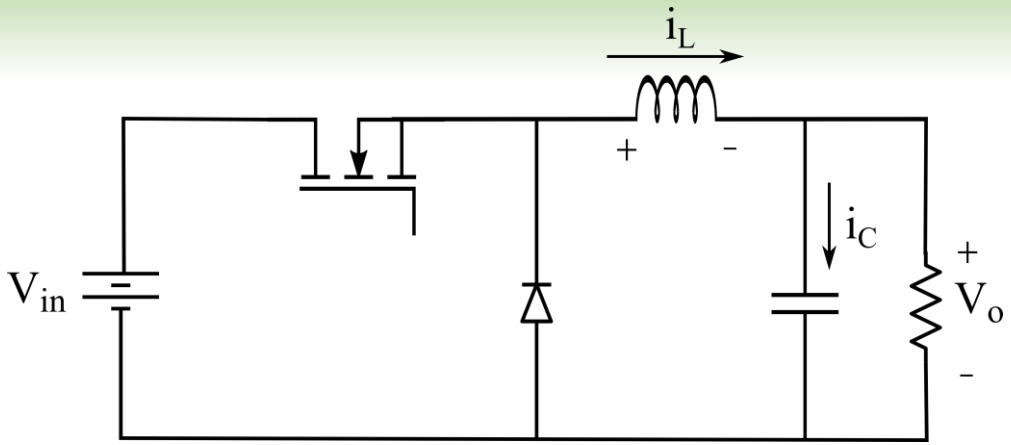
Non idealities

- There are several non idealities in Buck converter. Inductor resistance, diode and switch drops are some of them.
- Gain of a buck converter with inductor resistance is: $G = \left[\frac{D}{1+\frac{r_L}{R}}\right]$
  - When switch is on
    - $L\frac{d}{dt}i_L = (V_{in} - V_o - i_L r_L)$
  - When switch is off
    - $L\frac{d}{dt}i_L = -V_o - i_L r_L$
  - Apply volt second balance and assume $i_L = I_L$



Efficiency

- The actual efficiency of a buck converter is
- $\eta = \left[1 - \frac{V_{sw}}{V_{in}} - \frac{V_{diode}(1-d)}{dV_{in}}\right]\left[\frac{R}{R+r_L}\right]$



- Clearly, if the non-idealities are less in magnitude, then efficiency is higher. If $V_{sw} \ll V_{in}$, then efficiency is higher.
- The steady state equations and other non-idealities put a restriction on the range of duty cycle of the buck converter. A buck converter operating with 10V input and 1.5V output is not operating efficiently.

Other converters

- The same analysis can be extended to other converters like boost, buck-boost or Cuk converter.
  - Volt second balance on L and C gives duty ratio
  - Inductor current and capacitor voltage expressions
  - Design of L and C
  - Practical efficiency expression gives limits of operation

